

Student Researchers Breaking Barriers: Exploring How STEM Can Drive Education for Sustainability in UK Higher Education

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1 Introduction

- ❖ Education for sustainable development (ESD) plays a crucial role in equipping learners with the knowledge, skills, and attitudes needed to create a sustainable future.
- ❖ To explore how STEM subjects contribute to addressing Sustainable Development Goals (SDGs) in higher education, two undergraduate student researchers from QMUL and UCL took a lead role in driving the focus on:
 - RQ1: What are the challenges and opportunities for integrating ESD in engineering?
 - RQ2: What pedagogical strategies are most effective for helping students connect with traditionally inaccessible STEM areas?

3 SDG Module Mapping

- ❖ A systematic keyword analysis of 108 module descriptions was conducted to assess the extent to which SDGs were embedded within postgraduate engineering modules at QMUL and UCL.
- ❖ Keyword Search for SDGs: Module descriptions and tagged were analysed for alignment with SDGs using keywords like "sustainability," "climate," "resilience," and "equity." Findings were recorded and categorised by university, faculty, and department with multi-SDG modules noted and the extent of SDG integration explored.

Distribution of SDGs Across Mapped Courses

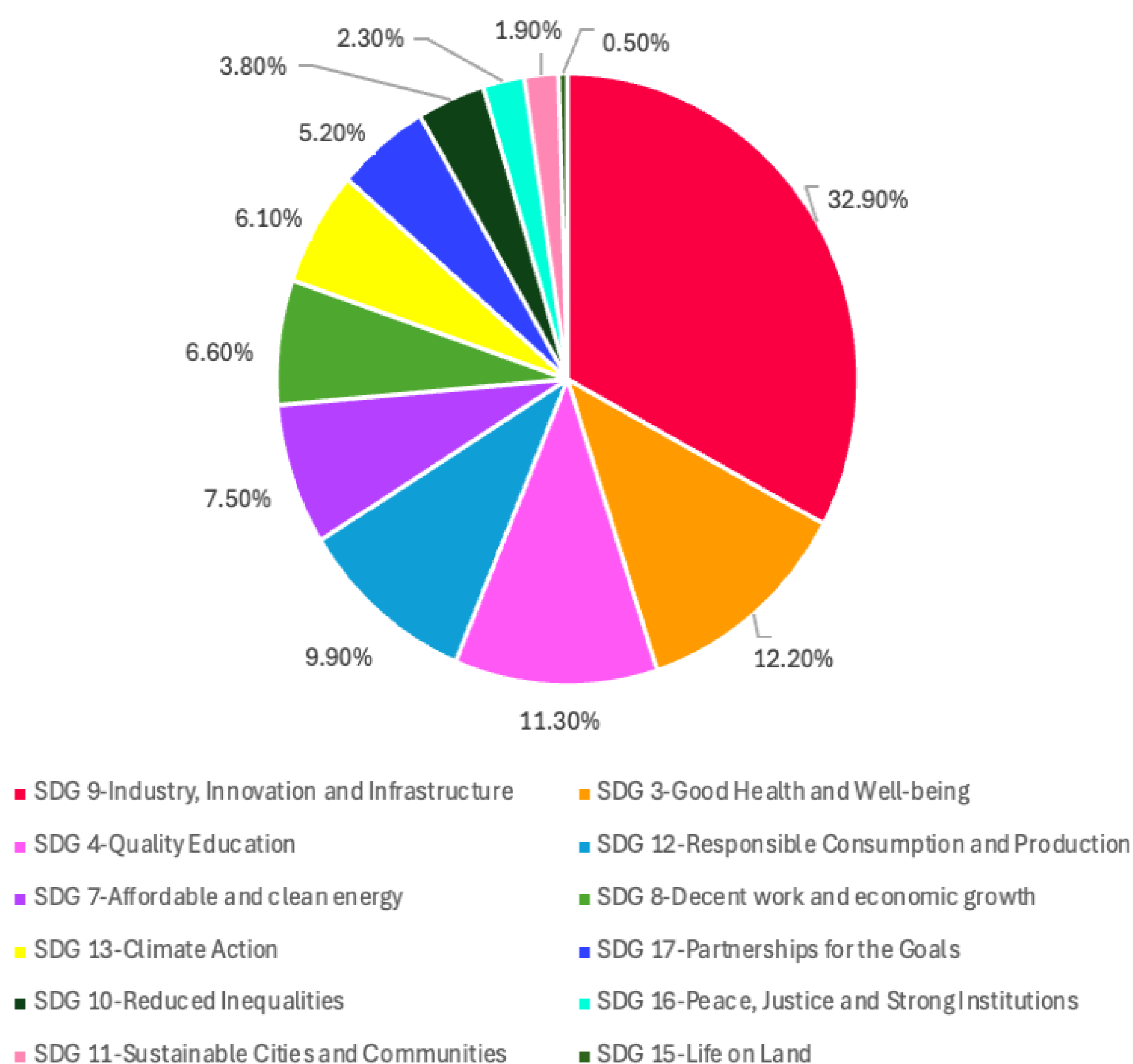


Figure 1: Distribution of SDGs across the mapped courses.

2 Literature Review - Innovative pedagogical strategies for embedding ESD in engineering education

- ❖ Rather than a bolt-on approach, literature suggests that universities adopt integration of strategies such as **Community-based learning and teaching** that combines experiential learning processes directed towards community service [1].
- ❖ Pedagogies that transcend disciplinary boundaries such as **interdisciplinary learning that engage with socio-scientific issues** can illuminate the connection of sustainability issues [2].

Modules by Number of SDGs Covered

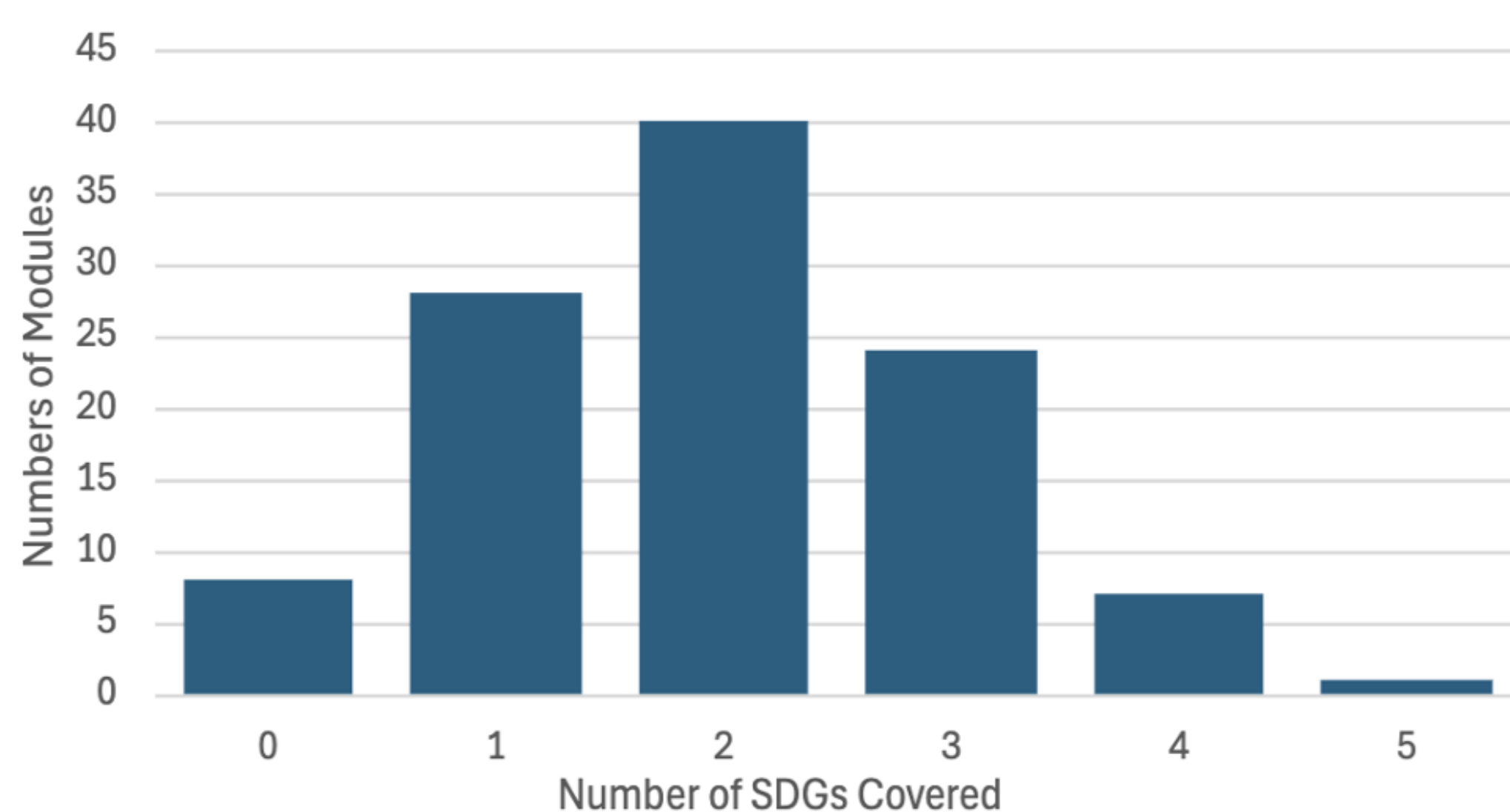


Figure 2: Modules by number of SDGs covered across the two institutions.

4 Qualitative Findings

- ❖ Existing **emphasis on SDG 9** alone accounts for **32.9%** of all SDG integration, which **reflects a proactive alignment of engineering curricula with industry needs** and infrastructural challenges, for students to engage with innovation and real-world problem-solving.
- ❖ Majority of modules (**63% of the dataset**) covered **primarily 1-2 SDGs**, suggesting a narrow approach to embedding sustainability. Modules extensively integrating 4+ SDGs were notably scarce with **less than 10 modules integrating over 4 SDGs**.

5 Conclusion

- ❖ **SDG 7, 13 and 11** directly related to environmental sustainability **were underrepresented**. This is particularly concerning given the core role engineers play in shaping built and environmental systems.
- ❖ **SDG integration remains selective, favouring industry goals over broader environmental and social priorities**. Limited interdisciplinary engagement also suggests missed opportunities for preparing graduates to address sustainability challenges.

6 References

- [1] Delaney, E., Liu, W., 2023, *Postgraduate design education and sustainability—An investigation into the current state of higher education and the challenges of educating for sustainability*. Department of Engineering, King's College London.
- [2] Hogan, D., & O'Flaherty, J. (2021). Addressing Education for Sustainable Development in the Teaching of Science: The Case of a Biological Sciences Teacher Education Program. *Sustainability*, 13(21), 12028.